

DUBBO, Australia

WMO: 957190

Lat: 32.2206S

Lon: 148.5753E

Elev: 285

StdP: 97.95

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.8	0.7	-6.2	2.3	20.7	-4.2	2.7	17.1	10.1	13.7	9.1	13.8	2.1	140	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.4	38.1	19.6	36.1	19.3	34.4	19.1	22.6	30.5	21.9	29.7	21.2	29.1	5.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	16.1	24.1	19.9	15.1	23.8	19.0	14.2	23.7	68.1	30.1	65.3	29.9	62.7	29.1	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.3	8.3	-3.3	41.5	0.9	2.0	-4.0	42.9	-4.5	44.1	-5.0	45.3	-5.7	46.8		
			-3.6	24.2	1.0	1.1	-4.3	25.0	-4.9	25.6	-5.5	26.2	-6.2	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	25.9	24.9	22.1	17.9	13.1	10.3	9.2	10.2	13.7	17.3	21.0	23.7
	DBStd	6.63	3.54	3.28	3.16	3.20	2.94	2.61	2.41	2.57	3.13	3.68	3.98	3.41
	HDD10.0	115	0	0	0	1	7	28	45	30	4	1	0	0
	HDD18.3	1222	1	1	7	45	163	242	284	253	143	64	17	3
	CDD10.0	2811	492	418	374	236	103	37	19	34	114	228	331	425
	CDD18.3	878	234	185	122	31	1	0	0	0	4	33	98	169
	CDH23.3	10468	2884	2025	1208	341	14	0	0	3	122	525	1242	2104
CDH26.7	5078	1611	1034	495	76	1	0	0	0	24	183	581	1074	
Wind	WSAvg	4.3	5.0	4.9	4.5	4.0	3.6	3.8	3.6	3.8	4.1	4.5	4.8	4.8
Precipitation	PrecAvg	617	70	52	47	40	54	38	49	50	49	57	52	53
	PrecMax	1172	220	268	189	292	221	107	138	144	159	210	148	115
	PrecMin	329	1	4	0	0	2	1	2	0	1	3	0	1
	PrecStd	185	55	50	43	55	42	26	32	31	38	41	39	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	40.6	35.8	31.1	25.0	20.8	20.1	23.2	29.2	33.2	39.0	39.0
		MCWB	20.5	21.3	19.2	17.5	15.6	13.6	11.9	13.1	14.7	16.6	20.0	18.6
	2%	DB	38.4	37.1	33.3	28.8	23.1	19.0	18.2	20.9	26.6	30.9	35.2	36.2
		MCWB	20.0	20.2	18.7	16.7	14.7	12.9	11.4	12.4	14.3	16.1	17.8	18.6
	5%	DB	36.3	34.8	31.4	27.0	21.7	17.5	16.8	19.0	24.2	28.8	32.4	34.4
		MCWB	19.8	19.8	18.5	16.1	13.8	12.0	11.1	11.6	13.8	15.8	17.4	18.5
	10%	DB	34.3	32.4	29.5	25.2	19.9	16.0	15.2	17.1	22.0	26.3	30.0	32.4
		MCWB	19.8	19.5	18.2	15.6	13.1	11.4	10.4	10.9	13.3	15.4	17.1	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	24.0	22.8	19.6	17.5	15.2	15.4	15.4	17.9	19.9	21.9	22.6
		MCDB	31.7	32.2	28.9	25.2	21.6	17.7	17.3	19.2	24.4	27.0	31.0	30.1
	2%	WB	22.5	22.7	21.4	18.3	16.1	14.1	13.3	14.0	16.3	18.7	20.6	21.4
		MCDB	31.4	30.2	27.3	24.8	20.0	16.6	15.7	18.0	22.5	26.1	29.1	29.2
	5%	WB	21.7	21.9	20.3	17.4	15.0	13.1	12.0	12.7	15.2	17.6	19.4	20.6
		MCDB	30.7	29.0	27.1	23.9	19.5	16.1	15.2	17.2	20.9	24.6	27.4	28.5
	10%	WB	21.1	21.1	19.6	16.5	14.1	12.2	11.0	11.6	14.2	16.5	18.4	19.8
		MCDB	29.8	28.6	26.3	22.9	18.8	15.3	14.1	16.3	19.7	23.4	26.1	27.9
Mean Daily Temperature Range	5% DB	MDBR	14.4	13.6	13.5	13.8	13.0	11.5	12.2	13.6	14.9	15.1	14.6	14.9
		MCDBR	16.9	16.4	15.8	15.5	14.8	12.9	14.2	16.5	18.7	18.6	18.1	17.5
		MCWBR	5.2	5.2	5.9	6.4	7.3	7.4	8.4	9.1	8.7	7.7	6.4	6.1
	5% WB	MCDBR	14.2	13.0	12.7	12.8	11.5	10.3	11.0	13.1	14.1	14.8	14.1	14.0
		MCWBR	5.2	4.9	5.6	5.8	6.3	6.5	7.2	8.0	7.6	7.2	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.352	0.350	0.324	0.306	0.289	0.285	0.280	0.288	0.318	0.329	0.348	0.342	
	taud	2.506	2.534	2.595	2.607	2.607	2.622	2.623	2.581	2.479	2.483	2.480	2.523	
	Ebn at Noon	987	969	961	928	896	875	899	935	951	978	984	1000	
	Edh at Noon	114	107	94	83	74	68	71	84	104	111	116	113	
All-Sky Solar Radiation	RadAvg	7.65	6.74	5.69	4.55	3.42	2.66	3.02	3.95	5.34	6.45	7.09	7.73	
	RadStd	0.40	0.49	0.41	0.30	0.19	0.20	0.18	0.26	0.35	0.48	0.67	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-89	N/A	N/A	
Regional (13 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-88	+191	+112	

Nomenclature: See separate page